Tetiana Parshakova

Research Fellow
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Research Interests

Efficient algorithms for large-scale problems using techniques from optimization, discrete mathematics, and statistics

- optimization for large language model pretraining, mid-training, and post-training
- numerical and randomized linear algebra
- large-scale and distributed convex optimization
- low-rank and structured optimization

Education

Stanford University — USA
Ph.D., Computational Mathematics. Advisor: Prof. Stephen Boyd — 2019 – 2024
Thesis: *Multilevel Low Rank Matrices and Applications*

Stanford University — USA
M.Sc., Computational Mathematics. Advisor: Prof. Stephen Boyd — 2019 – 2022

KAIST — South Korea
M.Sc., Electrical Engineering — 2017 – 2019
Thesis: *Latent Question Interpretation: Parameter Adaptation Using Interpretation Policy*

KAIST — South Korea
B.Sc., Industrial Design — 2012 – 2017
Thesis: *UMorph: Self-Change Tracker to Reflect Yourself to the Future and Past*

Positions

Flatiron Institute, CCM — USA
Research Fellow. Advisor: Robert Gower — 2025 –

- design and analysis of optimization algorithms for training large language models

Amazon, SCOT — USA
Postdoctoral Scientist. Advisor: Garrett van Ryzin — 2024 – 2025

- unified framework for empirically accelerated ADMM
- experimentation under constraints

Google Research, Brain Robotics — USA
Student Researcher — 2022

- message-passing and tree-based algorithms for fast graph field integration
- generalization of the fast multipole method to discretized manifolds

Apple, XDG — USA
Machine Learning Research Intern — 2020, 2021

- accelerating neural-network training using Hessian-vector products
- constructive methods for neural networks on elementary functions

Naver Labs Europe, NLP Group — France
Machine Learning Researcher — 2019

- data-efficient learning via energy-based models combined with autoregressive models

Patents

Naver Corp 2024
Methods and systems for producing neural sequential models US12086708B2
TP, M. Dymetman, J.-M. Andréoli

Apple Inc 2023
Interpolation method and apparatus for arithmetic functions US11636176B2
W. C. Athas, Z. M. Nadeem, **TP**

Publications

TP, R. M. Gower, P. Orenstein. *A Statistical Viewpoint on Adam, Shampoo and Muon.* arXiv:2610.24839, 2026
parshakova/matrix_normal_optimizers

J. Ko, **TP**, D. Cai, R. M. Gower. *SoftServe: A Scalable Quasi-Newton Method for Deep Learning.* arXiv:2610.02182, 2026
joohwanko/SoftServe

N. Ghosh, **TP**, R. M. Gower. *PoLoRA: a preconditioned orthogonalized LoRA optimizer.* arXiv:2607.17620, 2026
nikhilgsh/polora

TP, A. Khaled, M. Crawshaw, G. Garrigos, R. M. Gower. *Muon does not converge on convex Lipschitz functions.* NeurIPS (to appear), 2026

TP, Y. Bai, G. van Ryzin, S. Boyd. *Multiple approximate-response agents (MARA): fast near-optimal primal recovery for distributed optimization.* arXiv:2503.12221, 2025
cvxgrp/mra_precovery

TP, T. Hastie, S. Boyd. *Fitting multilevel factor models.* SIAM Journal on Matrix Analysis and Applications, 2025
cvxgrp/multilevel_factor_model

S. Boyd, **TP**, E. K. Ryu, J. J. Suh. *Optimization algorithm design via electric circuits.* NeurIPS (Spotlight), 2024
cvxgrp/optimization_via_circuits

TP. *Multilevel low rank matrices and applications.* Ph.D. thesis, Stanford University, 2024

TP, T. Hastie, E. Darve, S. Boyd. *Factor fitting, rank allocation, and partitioning in multilevel low rank matrices.* Optimization, Discrete Math., and Applications to Data Sciences, Springer Optimization and Its Applications, vol. 220, 2024
cvxgrp/mlr_fitting

K. Choromanski, A. Sehanobish, H. Lin, Y. Zhao, E. Berger, **TP**, et al. *Efficient graph field integrators meet point clouds.* ICML, 2023
topographers/efficient_graph_algorithms

TP, F. Zhang, S. Boyd. *Implementation of an oracle-structured bundle method for distributed optimization.* Optimization and Engineering, 2023
cvxgrp/OSBDO

TP, M. Dymetman, J.-M. Andreoli. *Distributional policies for energy-based sequential models.* NeurIPS Optimization Foundations of RL Workshop, 2019

TP, J.-M. Andreoli, M. Dymetman. *Global autoregressive models for data-efficient sequence learning.* CoNLL (ACL), 2019
parshakova/GAMS-for-Data-Efficient-Learning

TP, F. Rameau, A. Serdega, I. S. Kweon, D.-S. Kim. *Latent question interpretation through variational adaptation.* IEEE/ACM Trans. on Audio, Speech, and Language Processing, 2019

TP, D.-S. Kim. *Latent question interpretation through parameter adaptation using stochastic neuron.* ICML Workshop MRC, 2018

TP, D. Saakes. *UMorph: Self-change tracker to reflect yourself to the future and past.* ACM DIS Companion, 2018

TP, M. Cho, A. Cassinelli, D. Saakes. *Furniture that learns to move itself.* ACM CHI Extended Abstracts, 2017

TP, M. Cho, A. Cassinelli, D. Saakes. *Ratchair: Furniture learns to move itself with vibration.* ACM SIGGRAPH Emerging Technologies, 2016

Honors & Awards

2023	ICML Grant Award (declined)
2019 – 2022	Oliger Memorial Fellowship — Ph.D. stipend at Stanford
2018	Qualcomm-KAIST Innovation Awards — graduate paper competition
2017	Featured on Discovery Daily Planet Canada — *Ratchair* demonstration
2017	KAIST Breakthroughs Newsletter — *Furniture that learns to move itself*
2017	Excellence Award for Bachelor's Thesis — *UMorph*
2016	First Prize, Qualcomm-KAIST Innovation Awards — embedded systems competition
2016	SIGGRAPH Emerging Technologies, DC EXPO Special Prize — *Ratchair*
2016	Undergraduate Research Program Excellence Award
2017 – 2019	International Student Scholarship — M.Sc. stipend at KAIST
2012 – 2016	International Student Scholarship — B.Sc. stipend at KAIST
2009, 2012	Kyiv Capital Mathematics Olympiad — Bronze
2009	Regional Mathematics Olympiad — Silver
2008	Volyn Regional Mathematics Olympiad — Gold

Invited Talks

06/2026	*Structured Gradient Methods for Deep Learning*	NY CERN Alumni & Flatiron	New York
04/2026	*Muon Does Not Converge on Convex Lipschitz Functions*	ISL Colloquium	Stanford
04/2026	*Muon Does Not Converge on Convex Lipschitz Functions*	Apple XDG	Cupertino
10/2025	*Optimization Algorithm Design via Electric Circuits*	INFORMS	Atlanta
10/2025	*Optimization Algorithm Design via Electric Circuits*	Alg-ML Seminar	Princeton
10/2024	*Price Directed Distributed Optimization and Primal Recovery*	INFORMS	Seattle
05/2024	*Multilevel Low Rank Matrices and Applications*	Optimization Lunch	Stanford
05/2024	*Multilevel Low Rank Matrices and Applications*	Amazon SCOT	New York
01/2024	*Multilevel Low Rank Matrices and Applications*	van Dijk Lab	Yale
12/2023	*Multilevel Low Rank Matrices and Applications*	Krishnaswamy Lab	Yale
09/2022	*Fast Graph Field Integrators for Robotics & Beyond*	Google Brain	New York
01/2019	*Latent Question Interpretation*	Naver Labs Europe	Meylan
07/2018	*Latent Question Interpretation Through Parameter Adaptation*	ICML Workshop	Stockholm

Teaching

Course Assistant, Stanford University — USA
EE364a — Convex Optimization — 2023, 2024

Tutor, KAIST EE Co-op Program — South Korea
Undergraduate ML, NLP, and TensorFlow — 2018

Tutor, Science and English Camps — South Korea
Science-competition preparation; English — 2017, 2018

Teaching Assistant, KAIST — South Korea
Intro to Philosophy; English Short Stories; Philosophy of Math.; Logic & AI — 2015 – 2016

Community

NeurIPS Workshop Organizer — USA
ScaleOPT: GPU-Accelerated and Scalable Optimization — 2025
With P. Nobel, F. Zhang, M. Schaller, A. Amice, T. Marcucci, S. Boyd

Reviewer
JMLR 2026; Optimization and Engineering 2026; NeurIPS 2025; IEEE Conference on Decision and Control 2025; Energy-Based Models Workshop @ ICLR 2021

Organizer of KAIST EE Promotion in Ukraine — Ukraine / South Korea
Organized EE Visit Camp; recruited students — 2017